

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 46

Level 3 — Advanced · Logical Reasoning II

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 3 — Advanced	Topic Focus	Logical Reasoning II — Deduction & Counting Puzzles
Questions	20	Time Allowed	40 minutes
Total Marks	20	Worksheet No.	46 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1–5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 46 is pure problem-solving: overlapping-group (Venn) counting, logical deduction (the wrong-labels box, all-roses-are-flowers), the classic frog-in-the-well, a slow clock, handshake/match counting (pairs), and both-ends ranking. These Achievers-style puzzles reward drawing a picture and reasoning step by step — the heart of olympiad mathematics.

Learning Objectives

- Solve two-set overlap problems (both / only / neither).
- Make valid logical deductions and reject what is not guaranteed.
- Count pairs for lines, matches and handshakes.
- Solve classic puzzles: frog-in-well, slow clock, wrong labels.
- Combine ranking clues and set up simple 'difference' equations.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

- Q1.** In a class, 18 students like cricket, 15 like football, and 8 like BOTH. How many students like cricket ONLY?
- (A) 10 (B) 8
(C) 7 (D) 18
- Q2.** (Same class.) How many like football ONLY?
- (A) 7 (B) 8
(C) 15 (D) 23
- Q3.** 20 children were asked about juice and milk. 12 like juice, 9 like milk, and 3 like NEITHER. How many like BOTH?
- (A) 3 (B) 4
(C) 5 (D) 6
- Q4.** Five friends sit in a row. Reena is in the middle. Sita is at the far left. Tara is to Reena's right. Who is at the FAR RIGHT cannot be Tara, so the far-right person is among the other two. If Uma is immediately right of Tara, who is at the far right?
- (A) Sita (B) Uma
(C) Tara (D) Reena
- Q5.** A is taller than B. C is shorter than B. D is taller than A. Who is the TALLEST?
- (A) A (B) B
(C) C (D) D
- Q6.** If all ROSES are FLOWERS, and some FLOWERS are RED, which MUST be true?
- (A) All roses are red (B) Roses are flowers
(C) All flowers are roses (D) No rose is red
- Q7.** Today is Sunday. Riya's birthday is in 30 days. On which day will her birthday fall?
- (A) Monday (B) Tuesday
(C) Wednesday (D) Thursday
- Q8.** In how many ways can the letters **C, A, T** be arranged in a row?
- (A) 3 (B) 6
(C) 9 (D) 12
- Q9.** There are 5 points on a circle. How many straight lines can be drawn joining any two of them?
- (A) 5 (B) 8
(C) 10 (D) 20



Q10. A clock loses 2 minutes every hour. If set correctly at 12:00 noon, what will it show at the REAL time of 6:00 p.m.?

- (A) 5:48 p.m. (B) 5:50 p.m.
(C) 5:52 p.m. (D) 5:46 p.m.

Q11. Pinky is older than Queen but younger than Raja. Sunil is younger than Queen. Who is the YOUNGEST?

- (A) Pinky (B) Queen
(C) Raja (D) Sunil

Q12. A number is greater than 30, less than 40, and divisible by both 2 and 3. The number is:

- (A) 32 (B) 34
(C) 36 (D) 38

Q13. In a row of 7 children, Mina is 3rd from the left. Nita is 2nd from the right. How many children are BETWEEN Mina and Nita?

- (A) 1 (B) 2
(C) 3 (D) 4

Q14. If you fold a square piece of paper in half twice, then cut off one corner, how many holes are in the paper when unfolded? (Corner at the centre)

- (A) 1 (B) 2
(C) 3 (D) 4

Q15. Three boxes are labelled Apples, Oranges, Mixed — but ALL labels are wrong. You pick one fruit from the box labelled 'Mixed' and it is an apple. What does this box REALLY contain?

- (A) Apples (B) Oranges
(C) Mixed (D) Cannot tell

Q16. A frog climbs a 10 m well. Each day it climbs 3 m, but slips back 1 m at night (net 2 m a day). On the day it reaches the top it does NOT slip back. On which DAY does it reach the top?

- (A) Day 4 (B) Day 5
(C) Day 6 (D) Day 7

Q17. Rahul ranks 5th from the top and 18th from the bottom in his class. How many students are in the class?

- (A) 21 (B) 22
(C) 23 (D) 24

Q18. If MONDAY is written using number codes where M=13, O=15 ..., what is the code-sum of the FIRST and LAST letters of MONDAY?

- (A) 38 (B) 37
(C) 39 (D) 40



Q19. A basket has red and green balls. There are 3 more red than green. If there are 15 balls in all, how many are GREEN?

(A) 6

(B) 7

(C) 8

(D) 9

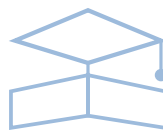
Q20. Four teams play each other ONCE in a tournament. How many matches are played in all?

(A) 4

(B) 6

(C) 8

(D) 12



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	A	A	B	B	D	B	B	B	C	A
Q	11	12	13	14	15	16	17	18	19	20
Ans	D	C	B	A	A	B	B	A	A	B

Step-by-Step Solutions

Q1. Correct answer: (A)

Cricket only = $18 - 8$ (the both) = **10** students.

Q2. Correct answer: (A)

Football only = $15 - 8 = 7$ students.



Q3. Correct answer: (B)

Like at least one = $20 - 3 = 17$. Both = $(12 + 9) - 17 = 21 - 17 = 4$.

Q4. Correct answer: (B)

Order: Sita _ Reena Tara Uma. Reena (middle, position 3), Tara (4), Uma (5) is at the far right: **Uma**.

Q5. Correct answer: (D)

Order: $D > A > B > C$. The tallest is **D**.

Q6. Correct answer: (B)

The only certain statement is that **roses are flowers** (given). The others are not guaranteed.

Q7. Correct answer: (B)

$30 \div 7 = 4$ weeks remainder 2. Two days after Sunday: Monday (1), **Tuesday** (2).

Q8. Correct answer: (B)

$3 \times 2 \times 1 = 6$ ways (CAT, CTA, ACT, ATC, TCA, TAC).



Q9. Correct answer: (C)

Each pair gives one line: $5 \times 4 \div 2 = 10$ lines.

Q10. Correct answer: (A)

From noon to 6 p.m. is 6 hours, losing 2 min each hour $\rightarrow$ 12 min behind. $6:00 - 12 \text{ min} = 5:48 \text{ p.m.}$

Q11. Correct answer: (D)

Order: Raja > Pinky > Queen > Sunil. The youngest is **Sunil**.

Q12. Correct answer: (C)

Divisible by 2 and 3 means divisible by 6. Between 30 and 40: **36**.

Q13. Correct answer: (B)

Mina is at position 3; Nita at position $7 - 2 + 1 = 6$. Between positions 3 and 6 are positions 4 and 5 $\rightarrow$ **2 children**.

Q14. Correct answer: (A)

Folding twice stacks 4 layers at the centre corner. One cut there removes the centre, leaving **1 hole** in the middle when unfolded.

**Q15. Correct answer: (A)**

Since the 'Mixed' label is wrong, the box is not mixed; it gave an apple, so it must contain only **apples**.

Q16. Correct answer: (B)

Height at end of each day (after slip): Day1 2 m, Day2 4 m, Day3 6 m, Day4 8 m. On Day5 it starts at 8 m and climbs 3 m to reach 11 m $\geq$ 10 m — it gets out. So the frog reaches the top on **Day 5**.

Q17. Correct answer: (B)

Total = $5 + 18 - 1 = 22$ **students**.

Q18. Correct answer: (A)

First letter M = 13, last letter Y = 25. Sum = $13 + 25 = 38$.

Q19. Correct answer: (A)

Green + (Green + 3) = 15 $\rightarrow 2 \times \text{Green} = 12 \rightarrow \text{Green} = 6$ (and red = 9).

Q20. Correct answer: (B)

Each pair plays once: $4 \times 3 \div 2 = 6$ **matches**.

